

Proof of Conjecture 3: Strict Monotonicity of ρ_p for the Hadamard Matrix

Statement

Theorem 1 (Conjecture 3). *For every integer $n > 6$ and every integer $p > 2$, there exists $M \in \mathbb{R}^{n \times n}$ such that*

$$\rho_p(M) < \rho_{p+1}(M),$$

where $\rho_r(M) := \inf \{ \|\Delta M \Delta^{-1}\|_{\ell_n^r \rightarrow \ell_n^r} \mid \Delta = \text{diag}(\delta_1, \dots, \delta_n), \delta_i > 0 \}$.

The proof proceeds in four steps. We first establish a zero-padding lemma (Section 1), then compute the exact ℓ^r -operator norm of the normalized 4×4 Hadamard matrix (Section 2), show that no diagonal scaling can improve this value (Section 3), and finally combine everything to prove Conjecture 3 (Section 4).

1 Zero-Padding Preserves ρ_r

Theorem 2 (Zero-padding preserves ρ_r). *Let $1 \leq k \leq n$, let $r \in [1, \infty)$, and let $B \in \mathbb{R}^{k \times k}$. Define*

$$M = \begin{pmatrix} B & 0 \\ 0 & 0_{n-k} \end{pmatrix} \in \mathbb{R}^{n \times n}.$$

Then

$$\rho_r^{(n)}(M) = \rho_r^{(k)}(B).$$

Proof. Norm identity. We first prove that for any $C \in \mathbb{R}^{k \times k}$,

$$\left\| \begin{pmatrix} C & 0 \\ 0 & 0_{n-k} \end{pmatrix} \right\|_{\ell_n^r \rightarrow \ell_n^r} = \|C\|_{\ell_k^r \rightarrow \ell_k^r}.$$

Write $x = (u, v) \in \mathbb{R}^n$ with $u \in \mathbb{R}^k$ and $v \in \mathbb{R}^{n-k}$. Then $\begin{pmatrix} C & 0 \\ 0 & 0 \end{pmatrix} x = (Cu, 0)$,

so $\left\| \begin{pmatrix} C & 0 \\ 0 & 0 \end{pmatrix} x \right\|_r = \|Cu\|_r$. Also,

$$\|x\|_r = (\|u\|_r^r + \|v\|_r^r)^{1/r} \geq (\|u\|_r^r)^{1/r} = \|u\|_r,$$

where the inequality uses $\|v\|_r^r \geq 0$ and the fact that $t \mapsto t^{1/r}$ is non-decreasing.

If $u = 0$, then $\left\| \begin{pmatrix} C & 0 \\ 0 & 0 \end{pmatrix} x \right\|_r = 0 \leq \|C\|_{\ell_k^r} \|x\|_r$, so the bound holds. If $u \neq 0$,

$$\frac{\left\| \begin{pmatrix} C & 0 \\ 0 & 0 \end{pmatrix} x \right\|_r}{\|x\|_r} = \frac{\|Cu\|_r}{(\|u\|_r^r + \|v\|_r^r)^{1/r}} \leq \frac{\|Cu\|_r}{\|u\|_r} \leq \|C\|_{\ell_k^r}.$$

In both cases the ratio is $\leq \|C\|_{\ell_k^r}$; taking the supremum over nonzero x gives $\leq \|C\|_{\ell_k^r}$. The reverse inequality is obtained by testing on $x = (u, 0)$:

$$\frac{\left\| \begin{pmatrix} C & 0 \\ 0 & 0 \end{pmatrix} x \right\|_r}{\|x\|_r} = \frac{\|Cu\|_r}{\|u\|_r},$$

so taking the supremum over nonzero u gives $\geq \|C\|_{\ell_k^r}$. This proves the norm identity.

Lower bound on $\rho_r^{(n)}(M)$. Let $\Delta = \text{diag}(\delta_1, \dots, \delta_n)$ with all $\delta_i > 0$. Write $D_1 = \text{diag}(\delta_1, \dots, \delta_k)$ and $D_2 = \text{diag}(\delta_{k+1}, \dots, \delta_n)$. Then

$$\Delta M \Delta^{-1} = \begin{pmatrix} D_1 & 0 \\ 0 & D_2 \end{pmatrix} \begin{pmatrix} B & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} D_1^{-1} & 0 \\ 0 & D_2^{-1} \end{pmatrix} = \begin{pmatrix} D_1 B D_1^{-1} & 0 \\ 0 & 0 \end{pmatrix}.$$

By the norm identity, $\|\Delta M \Delta^{-1}\|_{\ell_n^r} = \|D_1 B D_1^{-1}\|_{\ell_k^r} \geq \rho_r^{(k)}(B)$. Taking the infimum over all Δ gives $\rho_r^{(n)}(M) \geq \rho_r^{(k)}(B)$.

Upper bound on $\rho_r^{(n)}(M)$. Let $\epsilon > 0$. Choose $D_1 \in \mathbb{R}^{k \times k}$ positive diagonal with $\|D_1 B D_1^{-1}\|_{\ell_k^r} \leq \rho_r^{(k)}(B) + \epsilon$. Extend to $\Delta = \text{diag}(D_1, I_{n-k}) \in \mathbb{R}^{n \times n}$. Then $\Delta M \Delta^{-1} = \text{diag}(D_1 B D_1^{-1}, 0)$, so by the norm identity, $\|\Delta M \Delta^{-1}\|_{\ell_n^r} = \|D_1 B D_1^{-1}\|_{\ell_k^r} \leq \rho_r^{(k)}(B) + \epsilon$. Hence $\rho_r^{(n)}(M) \leq \rho_r^{(k)}(B) + \epsilon$. Since $\epsilon > 0$ is arbitrary, $\rho_r^{(n)}(M) \leq \rho_r^{(k)}(B)$.

Combining both bounds, $\rho_r^{(n)}(M) = \rho_r^{(k)}(B)$. $\square$

2 Norm of the Normalized Hadamard Matrix

Let

$$H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 4}.$$

We verify $H^T H = I$. *Symmetry.* From the matrix display, $h_{ij} = h_{ji}$ for all i, j (compare the (i, j) and (j, i) entries in each of the six off-diagonal pairs), so $H = H^T$. *Row norms.* Each row has four entries of absolute value $1/2$, so $\sum_j h_{ij}^2 = 4 \cdot \frac{1}{4} = 1$ for each i . *Row inner products.* Denoting the ± 1 pattern

of row i by $(\varepsilon_{i1}, \varepsilon_{i2}, \varepsilon_{i3}, \varepsilon_{i4})$, the inner product of rows $i \neq i'$ is $\frac{1}{4} \sum_j \varepsilon_{ij} \varepsilon_{i'j}$. Checking all six distinct pairs:

$$\begin{aligned} \langle \text{row 1, row 2} \rangle &= \frac{1}{4}(1 \cdot 1 + 1 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1)) = \frac{1}{4}(1 + 1 - 1 - 1) = 0, \\ \langle \text{row 1, row 3} \rangle &= \frac{1}{4}(1 \cdot 1 + 1 \cdot (-1) + 1 \cdot 1 + 1 \cdot (-1)) = \frac{1}{4}(1 - 1 + 1 - 1) = 0, \\ \langle \text{row 1, row 4} \rangle &= \frac{1}{4}(1 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) + 1 \cdot 1) = \frac{1}{4}(1 - 1 - 1 + 1) = 0, \\ \langle \text{row 2, row 3} \rangle &= \frac{1}{4}(1 \cdot 1 + 1 \cdot (-1) + (-1) \cdot 1 + (-1) \cdot (-1)) = \frac{1}{4}(1 - 1 - 1 + 1) = 0, \\ \langle \text{row 2, row 4} \rangle &= \frac{1}{4}(1 \cdot 1 + 1 \cdot (-1) + (-1) \cdot (-1) + (-1) \cdot 1) = \frac{1}{4}(1 - 1 + 1 - 1) = 0, \\ \langle \text{row 3, row 4} \rangle &= \frac{1}{4}(1 \cdot 1 + (-1) \cdot (-1) + 1 \cdot (-1) + (-1) \cdot 1) = \frac{1}{4}(1 + 1 - 1 - 1) = 0. \end{aligned}$$

Hence the rows of H are orthonormal, so $HH^T = I$. Since $H = H^T$, we have $H^T H = HH^T = I$.

Theorem 3. For every $r \in [2, \infty)$,

$$\|H\|_{\ell_4^r \rightarrow \ell_4^r} = 4^{1/2-1/r}.$$

Proof. Write $\|H\|_r := \|H\|_{\ell_4^r \rightarrow \ell_4^r}$.

Upper bound. We use two standard norm-comparison inequalities on $\mathbb{R}^4$.

Inequality (i): $\|y\|_r \leq \|y\|_2$ for $r \geq 2$. Since $r \geq 2$, for any $y \in \mathbb{R}^4$,

$$\|y\|_r^r = \sum_{i=1}^4 |y_i|^r \leq \left(\max_i |y_i| \right)^{r-2} \sum_{i=1}^4 |y_i|^2 \leq \|y\|_2^{r-2} \cdot \|y\|_2^2 = \|y\|_2^r,$$

so $\|y\|_r \leq \|y\|_2$.

Inequality (ii): $\|y\|_2 \leq 4^{1/2-1/r} \|y\|_r$ for $r \geq 2$. Apply Hölder's inequality with exponents $r/2$ and $r/(r-2)$ (interpreting the case $r=2$ as equality):

$$\|y\|_2^2 = \sum_{i=1}^4 |y_i|^2 \leq \left(\sum_{i=1}^4 |y_i|^r \right)^{2/r} \left(\sum_{i=1}^4 1 \right)^{1-2/r} = \|y\|_r^2 \cdot 4^{1-2/r},$$

so $\|y\|_2 \leq 4^{1/2-1/r} \|y\|_r$.

Upper bound for $\|H\|_r$. For any nonzero $x \in \mathbb{R}^4$, using $H^T H = I$:

$$\frac{\|Hx\|_r}{\|x\|_r} \leq \frac{\|Hx\|_2}{\|x\|_r} = \frac{\|x\|_2}{\|x\|_r} \leq 4^{1/2-1/r}.$$

Hence $\|H\|_r \leq 4^{1/2-1/r}$.

Lower bound. Take $x = (1, 1, 1, 1)^T$. Then $Hx = (2, 0, 0, 0)^T$ (the first row of H sums to $4 \cdot \frac{1}{2} = 2$; every other row sums to 0). For any $r \in [2, \infty)$,

$$\frac{\|Hx\|_r}{\|x\|_r} = \frac{(2^r)^{1/r}}{(1+1+1+1)^{1/r}} = \frac{2}{4^{1/r}} = 4^{1/2-1/r}.$$

Hence $\|H\|_r \geq 4^{1/2-1/r}$.

Combining both bounds, $\|H\|_r = 4^{1/2-1/r}$ for all $r \in [2, \infty)$. $\square$

3 Diagonal Scalings Cannot Improve H

Theorem 4. For every $r \in [2, \infty)$,

$$\rho_r^{(4)}(H) = 4^{1/2-1/r},$$

where H is the normalized 4×4 Hadamard matrix of Section 2.

Proof. Let $q = r/(r-1)$ be the conjugate exponent, so $1/r + 1/q = 1$.

Upper bound. Choosing $\Delta = I_4$, we get $\rho_r^{(4)}(H) \leq \|I_4 H I_4^{-1}\|_r = \|H\|_r = 4^{1/2-1/r}$ by Theorem 3.

Row-norm lower bound (key lemma). For any matrix $A = (a_{ij}) \in \mathbb{R}^{n \times n}$ and any row index i ,

$$\|A\|_{\ell^r \rightarrow \ell^r} \geq \left(\sum_{j=1}^n |a_{ij}|^q \right)^{1/q}.$$

Proof. For any $x \in \mathbb{R}^n$, $\|Ax\|_r \geq |(Ax)_i| = |\sum_j a_{ij}x_j|$. By Hölder's inequality, $|\sum_j a_{ij}x_j| \leq (\sum_j |a_{ij}|^q)^{1/q} \|x\|_r$, so $\|A\|_r \geq \sup_{\|x\|_r=1} |\sum_j a_{ij}x_j|$. To prove equality (when $a^{(i)} \neq 0$), set $S := \sum_j |a_{ij}|^q > 0$ and

$$x_j := \frac{\text{sgn}(a_{ij})|a_{ij}|^{q-1}}{\|a^{(i)}\|_q^{q-1}}, \quad \text{where } \|a^{(i)}\|_q = S^{1/q}.$$

Then, using $(q-1)r = q$ (which holds since $1/r + 1/q = 1$):

$$\|x\|_r^r = \sum_j \frac{|a_{ij}|^{(q-1)r}}{\|a^{(i)}\|_q^{(q-1)r}} = \frac{\sum_j |a_{ij}|^q}{\|a^{(i)}\|_q^q} = \frac{S}{S} = 1,$$

and

$$\sum_j a_{ij}x_j = \frac{\sum_j |a_{ij}|^q}{\|a^{(i)}\|_q^{q-1}} = \frac{S}{S^{(q-1)/q}} = S^{1/q} = \|a^{(i)}\|_q.$$

Hence the supremum equals $\|a^{(i)}\|_q$, proving the lemma.

Lower bound for all Δ . Let $\Delta = \text{diag}(\delta_1, \delta_2, \delta_3, \delta_4)$ with $\delta_i > 0$, and set $C = \Delta H \Delta^{-1}$. Since $|h_{ij}| = 1/2$ for all i, j , we have $|c_{ij}| = \delta_i/(2\delta_j)$.

Choose m with $\delta_m = \max_i \delta_i$. Then $\delta_m/\delta_j \geq 1$ for all j , so every entry in row m of C satisfies $|c_{mj}| = \delta_m/(2\delta_j) \geq 1/2$. Applying the row-norm lower bound to row m :

$$\|C\|_r \geq \left(\sum_{j=1}^4 |c_{mj}|^q \right)^{1/q} \geq \left(\sum_{j=1}^4 \left(\frac{1}{2} \right)^q \right)^{1/q} = (4 \cdot 2^{-q})^{1/q} = \frac{4^{1/q}}{2}.$$

Since $1/q = 1 - 1/r$, we have $4^{1/q}/2 = 4^{1-1/r}/2 = 4^{1/2-1/r}$.

Therefore $\|\Delta H \Delta^{-1}\|_r \geq 4^{1/2-1/r}$ for every positive diagonal Δ . Taking the infimum gives $\rho_r^{(4)}(H) \geq 4^{1/2-1/r}$.

Combining both bounds, $\rho_r^{(4)}(H) = 4^{1/2-1/r}$. $\square$

4 Proof of Conjecture 3

Proof of Theorem 1. Let $n > 6$ and $p > 2$ be integers. Since $n \geq 7$, we have $n - 4 \geq 3 \geq 0$, so the block matrix

$$M = \text{diag}(H, 0_{n-4}) \in \mathbb{R}^{n \times n}$$

is well-defined, where H is the 4×4 normalized Hadamard matrix.

Computing $\rho_p^{(n)}(M)$. By Theorem 2 (zero-padding) with $k = 4$ and $B = H$:

$$\rho_p^{(n)}(M) = \rho_p^{(4)}(H).$$

Since $p \geq 3 > 2$, Theorem 4 applies with $r = p$:

$$\rho_p^{(4)}(H) = 4^{1/2-1/p}.$$

Computing $\rho_{p+1}^{(n)}(M)$. By Theorem 2 with $k = 4$ and $B = H$:

$$\rho_{p+1}^{(n)}(M) = \rho_{p+1}^{(4)}(H).$$

Since $p + 1 \geq 4 > 2$, Theorem 4 applies with $r = p + 1$:

$$\rho_{p+1}^{(4)}(H) = 4^{1/2-1/(p+1)}.$$

Comparison. Since $p \geq 3$, we have $p < p + 1$, so $1/p > 1/(p + 1)$, hence

$$\frac{1}{2} - \frac{1}{p} < \frac{1}{2} - \frac{1}{p+1}.$$

The function $x \mapsto 4^x$ is strictly increasing, so

$$\rho_p(M) = 4^{1/2-1/p} < 4^{1/2-1/(p+1)} = \rho_{p+1}(M).$$

This exhibits an explicit $n \times n$ matrix M with $\rho_p(M) < \rho_{p+1}(M)$, proving Conjecture 3. $\square$

References

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